

Chapter 3, Section 2

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4. *Mayer-Vietoris Sequence.* Let Y_1, Y_2 be two closed subsets of X . Then there is a long exact sequence of cohomology with supports

$$\begin{aligned} \cdots \longrightarrow H_{Y_1 \cap Y_2}^i(X, \mathcal{F}) \longrightarrow H_{Y_1}^i(X, \mathcal{F}) \oplus H_{Y_2}^i(X, \mathcal{F}) \longrightarrow H_{Y_1 \cup Y_2}^i(X, \mathcal{F}) \longrightarrow \\ \longrightarrow H_{Y_1 \cap Y_2}^{i+1}(X, \mathcal{F}) \longrightarrow \cdots \end{aligned}$$

Proof. There is an exact sequence of sheaves

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{H}_{Y_1 \cap Y_2}^0(\mathcal{F}) & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1 \cap Y_2} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{H}_{Y_1}^0(\mathcal{F}) \oplus \mathcal{H}_{Y_2}^0(\mathcal{F}) & \longrightarrow & \mathcal{F} \oplus \mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1} \oplus \mathcal{F}_{X-Y_2} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{H}_{Y_1 \cup Y_2}^0(\mathcal{F}) & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1 \cup Y_2} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

which induces the desired long sequence of cohomology with supports by (1.1A). $\square$

6. Let X be a Noetherian topological space, and let $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ be a direct system of injective sheaves of Abelian groups on X . Then $\varinjlim \mathcal{I}_\alpha$ is also injective.

Proof. We follow the hint. One direction is clear. Conversely, let $i : \mathcal{N} \rightarrow \mathcal{M}$ be an injective morphism of sheaves. By the proof of (2.7) we can write $\mathcal{N} = \varinjlim \mathcal{N}_\beta$ where $\mathcal{N}_\beta$ is generated by the sections on some open U_β , and similarly for $\mathcal{M} = \varinjlim \mathcal{M}_\beta$. Notice that we can assume $\mathcal{N}$ and $\mathcal{M}$ are defined over the same direct system so that they belong to the same Abelian category. Thus, the inclusion map $i : \mathcal{N} \rightarrow \mathcal{M}$ can be broken down into inclusion maps $i_\beta : \mathcal{N}_\beta \rightarrow \mathcal{M}_\beta$. A direct system of morphisms $\mathcal{N}_\beta \rightarrow \mathcal{M}$ induces the same inclusion morphism $\mathcal{N} = \varinjlim \mathcal{N}_\beta \rightarrow \mathcal{M}$, so we reduce to the case when

$\mathcal{N}$ and $\mathcal{M}$ are generated by a single section over some open set U . We have an exact sequence

$$\begin{array}{ccccccc}
 & 0 & & 0 & & & \\
 & \searrow & & \downarrow & & & \\
 & & \mathcal{K} & & & 0 & \\
 & & \downarrow & \searrow & & \downarrow & \\
 0 & \longrightarrow & \mathcal{R} & \longrightarrow & \mathbb{Z}_U & \longrightarrow & \mathcal{N} \longrightarrow 0 \\
 & & & & \searrow & & \downarrow \\
 & & & & & & \mathcal{M} \\
 & & & & & & \searrow \\
 & & & & & & 0
 \end{array}$$

where all the maps are natural, and $\mathcal{R}, \mathcal{K}$ are kernels of the quotients $\mathcal{N}, \mathcal{M}$, respectively. It is not hard to see from above that any $f : \mathcal{N} \rightarrow \mathcal{I}$ naturally extends to $\mathcal{M}$, which is what we wanted to show.

Next, we show any subsheaf $\mathcal{R} \subseteq \mathbb{Z}_U$ such that $\mathbb{Z}_U/\mathcal{R}$ is generated by a single section must be finitely generated. Indeed, fix some $x \in X$. Following the proof of (2.7), there exists some open neighborhood $x \in V \subseteq U$ such that $\mathcal{R}|_V \cong d \cdot \mathbb{Z}|_V$ for some positive integer d . Since X is noetherian, we can cover U by finite number of such V , say V_i for $i = 1, \dots, n$. Therefore, there is an exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \bigoplus_{i=1}^n d_i \cdot \mathbb{Z}_{V_i} \longrightarrow \bigoplus_{i,j,k} d_{ijk} \cdot \mathbb{Z}_{V_i \cap V_j \cap V_k}$$

where d_{ijk} is the minimum of d_i, d_j, d_k . The terms on the right are finitely generated. Thus, $\mathcal{R}$ is finitely generated, and any $\mathcal{R} \rightarrow \varinjlim \mathcal{I}_\alpha$ must factor through one of the $\mathcal{I}_\alpha$ (each generator s_i of $\mathcal{R}$ factors through one of the $\mathcal{I}_{\alpha_i}$, so take any $\beta > \alpha_i$, which exists by definition of a direct system). $\square$

7. Let S^1 be the circle (with its usual topology), and let $\mathbb{Z}$ be the constant sheaf $\mathbb{Z}$.

(a) Show that $H^1(S^1, \mathbb{Z}) = \mathbb{Z}$, using our definition of cohomology.

(b) Now let $\mathcal{R}$ be the sheaf of germs of continuous real-valued functions on S^1 . Show that $H^1(S^1, \mathcal{R}) = 0$.

Proof.

(a) We remark that cohomology commutes with colimits on paracompact Hausdorff spaces. In particular, the statements of (II, Ex. 1.11), (2.9) hold for S^1 . Let A, B be closed subsets of S^1 homeomorphic to the unit interval such that $A \cup B = S^1$ and $A \cap B = \{P, Q\}$ for two distinct points P, Q in S^1 . From now on, for any closed subset C of S^1 , denote $\mathbb{Z}_C = i_* \mathbb{Z}$, where $i : C \hookrightarrow S^1$ is the inclusion map and $\mathbb{Z}$ is the constant sheaf on C . Without ambiguity $\mathbb{Z}$ will denote the constant sheaf on the ambient space. We claim the following sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}_A \oplus \mathbb{Z}_B \xrightarrow{\tau} \mathbb{Z}_{A \cap B} \longrightarrow 0$$

defined by $\Delta(a) = (a, a)$ and $\tau(a, b) = b - a$ is exact. In particular, there exists natural maps $i_A^\#, i_B^\# : \mathbb{Z} \rightarrow \mathbb{Z}_A, \mathbb{Z}_B$ associated to the inclusion maps $i_A, i_B : A, B \hookrightarrow S^1$, so that $\Delta = (i_A^\#, i_B^\#)$. In the same way, the associated morphism of sheaves of the inclusion maps $j_A, j_B : A \cap B \rightarrow A, B$ ascends to naturally defined maps $j_A^\#, j_B^\# : \mathbb{Z}_A, \mathbb{Z}_B \rightarrow \mathbb{Z}_{A \cap B}$ in the form of a restriction morphism. Thus, $\tau = j_B^\# - j_A^\#$. Exactness can be checked at the level of stalks. Suppose $R \notin A \cap B$. Then either $R \in A$ or $R \in B$, say $R \in A$. Then the stalks are $(\mathbb{Z})_R = \mathbb{Z}$, $(\mathbb{Z}_A)_R = \mathbb{Z}$, $(\mathbb{Z}_B)_R = 0$, which is an exact sequence. If $R \in A \cap B$, then the stalks are $(\mathbb{Z})_R = \mathbb{Z}$, $(\mathbb{Z}_A)_R = \mathbb{Z}$, $(\mathbb{Z}_B)_R = \mathbb{Z}$, $\mathbb{Z}_{A \cap B} = \mathbb{Z}$ defined by Δ and τ , which is clearly exact. Hence, the sequence is exact at all points, so the sequence is exact.

Taking cohomology, we get a long exact sequence of cohomology groups

$$\begin{aligned}
 0 &\longrightarrow H^0(S^1, \mathbb{Z}) \longrightarrow H^0(S^1, \mathbb{Z}_A) \oplus H^0(S^1, \mathbb{Z}_B) \xrightarrow{\tau_0} H^0(S^1, \mathbb{Z}_{A \cap B}) \longrightarrow \\
 &\longrightarrow H^1(S^1, \mathbb{Z}) \longrightarrow H^1(S^1, \mathbb{Z}_A) \oplus H^1(S^1, \mathbb{Z}_B) \longrightarrow H^1(S^1, \mathbb{Z}_{A \cap B}) \longrightarrow \cdots
 \end{aligned}$$

where $H^i(\mathbb{Z}_A \oplus \mathbb{Z}_B) \cong H^i(\mathbb{Z}_A) \oplus H^i(\mathbb{Z}_B)$ by (2.9). By (2.10), we have

$$\begin{aligned} H^0(S^1, \mathbb{Z}), H^0(S^1, \mathbb{Z}_A), H^0(S^1, \mathbb{Z}_B) &= \mathbb{Z}, \\ H^0(\mathbb{Z}_{A \cap B}) &= \mathbb{Z} \oplus \mathbb{Z}, \\ H^1(\mathbb{Z}_{A \cap B}) &= 0 \end{aligned}$$

The first line follows from the fact that A, B, S^1 are all connected and locally connected. The intersection $A \cap B$ is a noetherian space of dimension zero with two irreducible components, namely the points P and Q , so its space of global sections is free of rank two. Lastly, $H^1(S^1, \mathbb{Z}_{A \cap B}) = H^1(A \cap B, \mathbb{Z}) = 0$ by (2.7). By exactness, we reduce to the following exact sequence

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} / \text{im } \tau_0 \longrightarrow H^1(S^1, \mathbb{Z}) \longrightarrow H^1(S^1, \mathbb{Z}_A) \oplus H^1(S^1, \mathbb{Z}_B).$$

The homomorphism τ_0 is defined by $\tau_0(a, b) = (b - a, b - a)$, which is the diagonal map. Thus, the term on the left is free of rank one. It remains to show $H^1(S^1, \mathbb{Z}_A) = H^1(S^1, \mathbb{Z}_B) = 0$. By (2.10), it suffices to show $H^1(A, \mathbb{Z}) = 0$.

From here, $\mathbb{Z}$ will denote the constant sheaf on A . Identifying A with the closed unit interval $[0, 1]$, we repeat the procedure above for A . Pick any $t \in (0, 1)$, say $t = 2^{-1}$. Then $X = [0, t]$ and $Y = [t, 1]$ cover A , so taking cohomology groups, we get a long exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(A, \mathbb{Z}) \longrightarrow H^0(A, \mathbb{Z}_X) \oplus H^0(A, \mathbb{Z}_Y) \longrightarrow H^0(A, \mathbb{Z}_{X \cap Y}) \longrightarrow \\ \longrightarrow H^1(A, \mathbb{Z}) \longrightarrow H^1(A, \mathbb{Z}_X) \oplus H^1(A, \mathbb{Z}_Y) \longrightarrow H^1(A, \mathbb{Z}_{X \cap Y}) \longrightarrow \dots \end{aligned}$$

Imitating the previous calculation, the first row is exact and $H^1(A, \mathbb{Z}_{X \cap Y}) = 0$ by (2.7) and (2.10). Thus, we reduce to the following exact sequence

$$0 \longrightarrow H^1(A, \mathbb{Z}) \longrightarrow H^1(A, \mathbb{Z}_X) \oplus H^1(A, \mathbb{Z}_Y) \longrightarrow 0.$$

Since $X \cong A, B$, $H^1(A, \mathbb{Z}_X), H^1(A, \mathbb{Z}_Y) \cong H^1(A, \mathbb{Z})$ by (2.10), which is possible if and only if $H^1(A, \mathbb{Z}) = 0$.

- (b) Let $\mathcal{M}$ be the sheaf of germs of measurable real-valued functions on S^1 modulo equivalence almost everywhere. It is clearly flasque, since for any measurable $f : V \rightarrow \mathbb{R}$ where $V \subseteq U \subseteq \mathbb{R}$ are open sets, the extension of f by zero on U is a measurable function. Thus, we have an exact sequence of sheaves

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{M} \longrightarrow \mathcal{M}/\mathcal{R} \longrightarrow 0.$$

Any measurable function on S^1 is continuous except for on a set of measure zero in S^1 , so any global section of $\mathcal{M}/\mathcal{R}$ is locally zero almost everywhere. Since we are considering equivalence classes of functions, where two functions are considered equal if they are equal except for on a measure zero set, we conclude that $\mathcal{M}/\mathcal{R}$ has no non-zero global section. Taking cohomology groups, we have $H^1(S^1, \mathcal{R}) = 0$.

□